

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 2 — Atoms, Molecules & Stoichiometry

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	2.1 Relative Masses · 2.2 The Mole & L · 2.3 Empirical/Molecular Formulas · 2.4 Reacting Volumes
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	232 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

2.1 Relative masses of atoms and molecules

— Ar and Mr definitions, unified atomic mass unit, carbon-12 standard, isotopic mass

2.2 The mole and the Avogadro constant— $n = m/M$, $N = n \times L$ calculations, moles of atoms, ions, subatomic particles, and Faraday constant**2.3 Formulas (empirical, molecular, structural)**— Combustion analysis ($C_xH_yO_z$), percentage composition by mass, hydrated salts ($CuSO_4 \cdot xH_2O$)**2.4 Reacting masses and volumes**— Acid-base & back titrations, gas molar volume, ideal gas $pV = nRT$, % yield and % atom economy

SUBTOPIC 2.1 — RELATIVE MASSES OF ATOMS & MOLECULES

1 Definitions of Relative Masses — 9701/22/M/J/21/Q1(a)

EASY · Syllabus 2.1

Accurate atomic and molecular masses are defined relative to the carbon-12 standard.

- (a) Define the term relative molecular mass, M_r .

[2]

- (b) Explain why the term relative formula mass is preferred over relative molecular mass for magnesium chloride, $MgCl_2$.

[1]

2 Calculating Relative Formula Masses — 9701/11/O/N/22/Q1

EASY · Syllabus 2.1

Use the following relative atomic mass values: H = 1.0, C = 12.0, N = 14.0, O = 16.0, Al = 27.0, S = 32.1, Fe = 55.8.

- (a) Calculate the relative formula mass of hydrated aluminium sulfate, $Al_2(SO_4)_3 \cdot 16H_2O$.

[2]

- (b) Calculate the relative formula mass of ammonium iron(II) sulfate, $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$.

[2]

3 High-Resolution Mass Measurement and Molecular Formula — 9701/21/M/J/23/Q1(c)

HARD · Syllabus 2.1

High-resolution mass spectrometry determines molecular masses to four decimal places using precise isotopic masses: $^1H = 1.0078$, $^{12}C = 12.0000$, $^{14}N = 14.0031$, $^{16}O = 15.9949$.

- (a) Both carbon monoxide, CO, and nitrogen gas, N_2 , have nominal integer relative molecular masses of 28. Calculate the precise molecular mass of CO and N_2 to four decimal places.

[2]

- (b) A gaseous compound has a molecular ion peak at $m/z = 44.0262$. Deduce whether the gas is propane (C_3H_8) or ethanal (CH_3CHO).

[2]

4 Isotopic Abundances and Weighted Average A_r — 9701/22/F/M/22/Q1(a)

EASY · Syllabus 2.1

Antimony, Sb ($Z = 51$), consists of two stable isotopes: ^{121}Sb (atomic mass 120.90 u) and ^{123}Sb (atomic mass 122.90 u). Natural antimony has $A_r = 121.76$.

- (a) State what is meant by the unified atomic mass unit, u. [1]

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- (b) Calculate the percentage abundance of each isotope in natural antimony. Give your answers to 1 decimal place. [3]

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5 Mass Differences Between Allotropes — 9701/12/M/J/22/Q2

EASY · Syllabus 2.1

Carbon exists as several allotropes, including diamond, graphite, and buckminsterfullerene (C₆₀).

- (a) State the relative molecular mass of buckminsterfullerene, C₆₀. [1]

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- (b) Explain why diamond does not have a defined relative molecular mass. [1]

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6 Evaluating Atomic Mass Standards Historically — 9701/22/O/N/20/Q1(a)

HARD · Syllabus 2.1

Prior to 1961, physicists calibrated atomic masses using oxygen-16 = 16.0000, while chemists used the natural isotopic mixture of oxygen = 16.0000.

- (a) Explain why the existence of ¹⁷O and ¹⁸O isotopes caused a discrepancy between the physical and chemical atomic mass scales. [2]

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- (b) Explain why adopting carbon-12 resolved this historical discrepancy universally. [2]

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7 Relative Formula Mass of Complex Coordination Compounds — 9701/21/M/J/24/Q1(b)

HARD · Syllabus 2.1

Cisplatin is an anticancer drug with the chemical formula Pt(NH₃)₂Cl₂. (Pt = 195.1, Cl = 35.5, N = 14.0, H = 1.0).

- (a) Calculate the relative molecular mass of cisplatin to one decimal place. [1]

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- (b) Calculate the percentage by mass of platinum in cisplatin. [2]

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8 Percentage Composition by Mass in Fertilizers — 9701/11/M/J/23/Q2

EASY · Syllabus 2.1

Ammonium nitrate, NH_4NO_3 , and urea, $(\text{NH}_2)_2\text{CO}$, are nitrogenous fertilizers. ($\text{H} = 1.0$, $\text{C} = 12.0$, $\text{N} = 14.0$, $\text{O} = 16.0$).

- (a) Calculate the percentage by mass of nitrogen in pure ammonium nitrate, NH_4NO_3 . [2]

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- (b) Calculate the percentage by mass of nitrogen in pure urea, $(\text{NH}_2)_2\text{CO}$. [2]

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- (c) Deduce which fertilizer provides a higher nitrogen mass per kilogram of pure compound. [1]

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9 Relative Masses of Isotopically Labelled Molecules — 9701/22/O/N/23/Q2(a)

HARD · Syllabus 2.1

Heavy methane is synthesized using carbon-13 and deuterium (2H , represented as D).

- (a) Calculate the relative molecular mass of standard methane, $^{12}\text{CH}_4$, and fully deuterated heavy methane, $^{13}\text{CD}_4$. [2]

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- (b) Calculate the percentage mass increase of $^{13}\text{CD}_4$ compared to $^{12}\text{CH}_4$. [2]

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10 Evaluating Hydration Mass in Epsom Salts — 9701/23/M/J/22/Q1(b)

EASY · Syllabus 2.1

Epsom salts consist of hydrated magnesium sulfate, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$. ($\text{Mg} = 24.3$, $\text{S} = 32.1$, $\text{O} = 16.0$, $\text{H} = 1.0$).

- (a) Calculate the formula mass of anhydrous MgSO_4 . [1]

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- (b) Given that water accounts for 51.2% of the mass of Epsom salts, deduce the integer value of x . [3]

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11 Relative Masses in Polymer Repeat Units — 9701/13/O/N/21/Q2

EASY · Syllabus 2.1

Poly(chloroethene), PVC, is formed by addition polymerisation of chloroethene, $\text{CH}_2=\text{CHCl}$. ($\text{C} = 12.0$, $\text{H} = 1.0$, $\text{Cl} = 35.5$).

- (a) State the relative formula mass of one repeat unit of PVC. [1]

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- (b) A sample of PVC has an average molecular mass of 187,500. Calculate the average number of monomer units in a polymer chain. [2]

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12 Distinguishing Isomers by Accurate Molecular Mass — 9701/22/F/M/24/Q1(b)

HARD · Syllabus 2.1

Consider the two structural isomers butane (C_4H_{10}) and 2-methylpropane (C_4H_{10}).

- (a) Explain why high-resolution mass spectrometry cannot be used to distinguish between butane and 2-methylpropane. [1]

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- (b) Suggest an analytical technique that could definitively differentiate the two isomers based on their mass fragmentation patterns. [2]

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SUBTOPIC 2.2 — THE MOLE & THE AVOGADRO CONSTANT

13 Defining the Mole and Avogadro Constant — 9701/22/M/J/22/Q2(a)

EASY · Syllabus 2.2

The mole is the SI unit for amount of substance. (Avogadro constant $L = 6.02 \times 10^{23} \text{ mol}^{-1}$).

- (a) Define the mole in terms of the Avogadro constant. [1]

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- (b) Calculate the number of molecules present in 3.60 g of water, H_2O . [2]

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- (c) Calculate the total number of atoms present in 3.60 g of water. [2]

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14 Counting Subatomic Particles in a Given Mass — 9701/11/M/J/21/Q2

HARD · Syllabus 2.2

A pure sample of magnesium metal has a mass of 1.216 g. (Ar: Mg = 24.31; atomic number $Z = 12$; $L = 6.02 \times 10^{23} \text{ mol}^{-1}$).

- (a) Calculate the amount in moles of magnesium atoms in the sample. [1]

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- (b) Calculate the total number of protons present in the 1.216 g sample of magnesium. [2]

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- (c) Calculate the total charge in coulombs carried by all the valence electrons in this sample. (Charge on electron $e = 1.60 \times 10^{-19}$ C). [3]

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15 Comparing Particle Quantities in Diverse Samples — 9701/12/O/N/23/Q1

EASY · Syllabus 2.2

Consider four separate containers each holding a different gas at standard conditions.

- (a) Which sample contains the greatest total number of atoms? [1]
- A 1.0 g of hydrogen gas, H_2
- B 1.0 g of helium gas, He
- C 1.0 g of methane gas, CH_4
- D 1.0 g of oxygen gas, O_2

- (b) Demonstrate by calculation why the option chosen in part (a) is correct. [2]

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16 Moles of Ions in Dissolved Salts — 9701/21/O/N/22/Q2(b)

EASY · Syllabus 2.2

Solid aluminium chloride, $AlCl_3$, dissolves completely in water to form hydrated ions. ($L = 6.02 \times 10^{23} \text{ mol}^{-1}$).

- (a) Write the equation for the dissolution of solid $AlCl_3$ in water. [1]

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- (b) Calculate the total number of chloride ions, Cl^- , present in 250 cm^3 of a $0.200 \text{ mol dm}^{-3}$ $AlCl_3$ solution. [3]

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17 Avogadro's Number and Mass of an Individual Atom — 9701/22/F/M/21/Q2(a)

HARD · Syllabus 2.2

Gold has a relative atomic mass of 197.0. A single atom of gold has mass m .

- (a) Calculate the absolute mass in grams of one atom of gold to three significant figures. ($L = 6.022 \times 10^{23} \text{ mol}^{-1}$). [2]

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- (b) A gold leaf has dimensions $5.0 \text{ cm} \times 5.0 \text{ cm}$ and a thickness of $1.0 \times 10^{-4} \text{ cm}$. The density of gold is 19.3 g cm^{-3} . Calculate the number of gold atoms in the leaf. [3]

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18 Moles and Faraday's Constant Relationship — 9701/23/M/J/24/Q1(b)

HARD · Syllabus 2.2

Faraday's constant F is the electric charge carried by one mole of electrons. ($e = 1.602 \times 10^{-19} \text{ C}$, $L = 6.022 \times 10^{23} \text{ mol}^{-1}$).

- (a) Show by calculation that Faraday's constant F is approximately $96,500 \text{ C mol}^{-1}$. [2]

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- (b) In an electrolysis experiment, a constant current of 2.50 A was passed through molten lead(II) bromide, PbBr_2 , for 30.0 minutes. Calculate the amount in moles of electrons transferred. [3]

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19 Determining Avogadro's Constant Experimentally — 9701/21/M/J/25/Q1(b)

HARD · Syllabus 2.2

Copper can be deposited during electrolysis according to: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$. In an experiment, a current of 0.800 A depositing 0.474 g of Cu ($A_r = 63.55$) in 1800 s was used to determine the Avogadro constant.

- (a) Calculate the amount in moles of copper atoms deposited. [1]

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- (b) Calculate the total electric charge passed and hence the number of electrons transferred, given that each electron carries $1.60 \times 10^{-19} \text{ C}$. [2]

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- (c) Calculate the experimental value of the Avogadro constant obtained from these data. [2]

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20 Moles of Gas and Molar Gas Volume — 9701/11/F/M/23/Q1

EASY · Syllabus 2.2

At room temperature and pressure (r.t.p.), one mole of any ideal gas occupies 24.0 dm^3 .

- (a) Calculate the volume in cm^3 occupied by 0.0250 mol of carbon dioxide gas at r.t.p. [2]

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- (b) Calculate the amount in moles of argon gas present in 360 cm^3 measured at r.t.p. [2]
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21 Mole Calculations Involving Solutions — 9701/12/M/J/24/Q2

EASY · Syllabus 2.2

A stock solution of sodium hydroxide, NaOH, is prepared by dissolving 8.00 g of NaOH ($M_r = 40.0$) in distilled water to make 500 cm^3 of solution.

- (a) Calculate the concentration of the NaOH solution in mol dm^3 . [2]
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- (b) Calculate the volume of this solution required to provide exactly 0.0100 mol of NaOH. [2]
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22 Number of Molecules in a Liquid Droplet — 9701/22/O/N/21/Q1(c)

HARD · Syllabus 2.2

A single drop of liquid ethanol, $\text{C}_2\text{H}_5\text{OH}$, has a volume of 0.050 cm^3 . The density of ethanol is 0.789 g cm^{-3} . ($C = 12.0$, $H = 1.0$, $O = 16.0$; $L = 6.02 \times 10^{23} \text{ mol}^{-1}$).

- (a) Calculate the mass in grams of the drop of ethanol. [1]
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- (b) Calculate the number of ethanol molecules present in this single droplet. [3]
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23 Limiting Reagent Concept from Molar Quantities — 9701/11/O/N/24/Q2

EASY · Syllabus 2.2

Consider the reaction: $2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$. A reaction mixture contains 0.50 mol of Al and 0.60 mol of Cl_2 .

- (a) Deduce which reactant is the limiting reagent. Justify your answer with a calculation. [2]
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- (b) Calculate the maximum amount in moles of AlCl_3 that can be formed. [1]
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24 Moles of Water in Hydrated Crystals — 9701/22/M/J/23/Q2(a)

EASY · Syllabus 2.2

A student dissolves 4.99 g of blue hydrated copper(II) sulfate crystals, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ ($M_r = 249.6$), in water. ($L = 6.02 \times 10^{23} \text{ mol}^{-1}$).

- (a) Calculate the amount in moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ dissolved. [1]

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- (b) Calculate the number of water molecules chemically bound in this 4.99 g sample. [2]

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SUBTOPIC 2.3 — EMPIRICAL & MOLECULAR FORMULAS

25 Thermal Gravimetric Analysis of Hydrated Crystals — 9701/22/M/J/23/Q2(b)-(d)

HARD · Syllabus 2.3

Fig. 25.1 shows the thermal gravimetric analysis (TGA) curve obtained when a 5.00 g sample of hydrated copper(II) sulfate, $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$, is heated progressively from 25 °C to 350 °C.

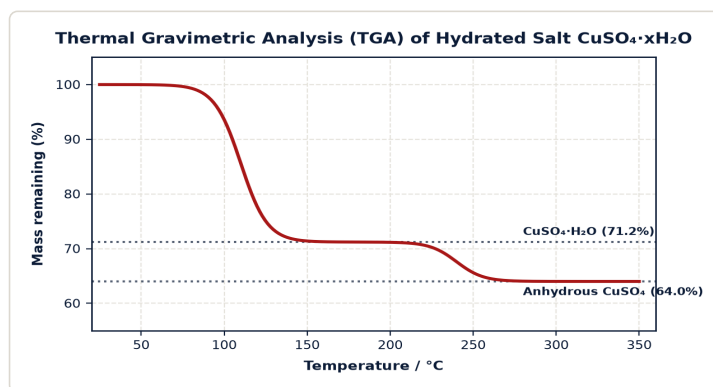


Fig. 25.1 Thermal gravimetric mass loss curve of $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$ against temperature

- (a) State the mass percentage remaining when all water of crystallisation has been expelled at 300 °C. [1]

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- (b) Calculate the formula mass of anhydrous CuSO_4 . ($\text{Cu} = 63.5$, $\text{S} = 32.1$, $\text{O} = 16.0$). [1]

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- (c) Use the mass loss data from Fig. 25.1 to deduce the value of x in the formula $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$. Show your working clearly. [3]

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26 Empirical Formula from Combustion Analysis — 9701/21/O/N/21/Q2(a)

HARD · Syllabus 2.3

Complete combustion of 2.16 g of an organic liquid compound J containing only carbon, hydrogen, and oxygen produced 4.40 g of carbon dioxide and 2.16 g of water. ($\text{C} = 12.0$, $\text{H} = 1.0$, $\text{O} = 16.0$).

- (a) Calculate the mass of carbon and the mass of hydrogen present in the 2.16 g sample of J. [2]

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- (b) Determine by calculation whether compound J contains oxygen, and if so, calculate its mass. [2]

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- (c) Deduce the empirical formula of compound J. [3]

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27 Empirical and Molecular Formula of an Ester — 9701/22/F/M/23/Q2(a)

EASY · Syllabus 2.3

A sweet-smelling ester K contains 58.8% carbon, 9.8% hydrogen, and 31.4% oxygen by mass. Compound K has a relative molecular mass of 102.0.

- (a) Define the term empirical formula. [1]

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- (b) Calculate the empirical formula of ester K. [3]

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- (c) Deduce the molecular formula of ester K. [1]

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28 Formula of a Hydrated Barium Salt — 9701/22/O/N/22/Q2(a)

EASY · Syllabus 2.3

A student heated 6.10 g of hydrated barium chloride crystals, $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$, in a crucible until constant mass was achieved. The mass of anhydrous BaCl_2 remaining was 5.20 g. (Ba = 137.3, Cl = 35.5, H = 1.0, O = 16.0).

- (a) Explain why the crucible was heated until constant mass was reached. [1]

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- (b) Calculate the mass of water expelled during heating. [1]

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- (c) Calculate the value of x in the formula $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$. [3]

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29 Combustion Analysis of a Hydrocarbon Gas — 9701/21/M/J/22/Q2(a)

HARD · Syllabus 2.3

A 20.0 cm^3 sample of a gaseous hydrocarbon C_xH_y was completely burned in 150 cm^3 of oxygen (an excess). After cooling to room temperature, the total gas volume was 110 cm^3 . Passing the gas through aqueous KOH reduced the volume to 30 cm^3 .

- (a) State the identity of the gas absorbed by aqueous potassium hydroxide, KOH. [1]

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- (b) Deduce the volume of carbon dioxide, CO_2 , produced during combustion. [1]

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- (c) Determine the volume of oxygen gas consumed in the reaction. [2]

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- (d) Deduce the molecular formula of the hydrocarbon C_xH_y . [3]

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30 Empirical Formula of an Iron Oxide — 9701/12/M/J/21/Q3

EASY · Syllabus 2.3

A sample of an oxide of iron contains 70.0% iron and 30.0% oxygen by mass. ($\text{Fe} = 55.8$, $\text{O} = 16.0$).

- (a) Calculate the empirical formula of the iron oxide. [3]

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- (b) State the oxidation state of iron in this oxide. [1]

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31 Formula of a Nitrogen Oxide from Decomposition — 9701/22/F/M/24/Q2(b)

HARD · Syllabus 2.3

When 50 cm^3 of a gaseous oxide of nitrogen N_xO_y decomposes completely, it yields 50 cm^3 of nitrogen gas (N_2) and 25 cm^3 of oxygen gas (O_2), all volumes measured under identical conditions of temperature and pressure.

- (a) State Avogadro's law regarding gas volumes. [1]
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- (b) Write a balanced equation for the decomposition in terms of x and y. [1]
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- (c) Deduce the molecular formula of N_xO_y . [2]
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32 Empirical Formula of a Chlorinated Alkane — 9701/11/O/N/23/Q3

EASY · Syllabus 2.3

A chloroalkane contains 24.2% carbon, 4.0% hydrogen, and 71.8% chlorine by mass. Its relative molecular mass is 99.0. (C = 12.0, H = 1.0, Cl = 35.5).

- (a) Calculate the empirical formula of the chloroalkane. [3]
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- (b) Determine its molecular formula. [1]
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33 Hydrated Sodium Carbonate Analysis — 9701/21/M/J/23/Q2(a)

HARD · Syllabus 2.3

Washing soda crystals have the formula $Na_2CO_3 \cdot 10H_2O$. On standing in dry air, they effloresce (lose water) to form a partially dehydrated salt $Na_2CO_3 \cdot xH_2O$. A 4.29 g sample of washing soda lost 2.43 g of water.

- (a) Calculate the amount in moles of Na_2CO_3 present in the original sample. (Na = 23.0, C = 12.0, O = 16.0, H = 1.0). [2]
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- (b) Calculate the amount in moles of water remaining in the partially dehydrated salt. [2]
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- (c) Deduce the value of x in the formula of the effloresced salt. [2]
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34 Formula of a Metal Nitrate from Oxide Residue — 9701/22/M/J/24/Q2(a)

HARD · Syllabus 2.3

Thermal decomposition of 3.40 g of an anhydrous Group 2 metal nitrate, $M(NO_3)_2$, produces 1.60 g of solid metal oxide, MO , along with nitrogen dioxide and oxygen gas.

- (a) Write a balanced equation for the thermal decomposition of $M(NO_3)_2$. [1]

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- (b) Set up an equation in terms of the atomic mass A_r of metal M to calculate A_r . [3]

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- (c) Identify metal M . [1]

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35 Structural vs Molecular Formulas of Isomers — 9701/13/M/J/22/Q2

EASY · Syllabus 2.3

An organic compound has the molecular formula C_3H_6O .

- (a) Draw the displayed formula of an aldehyde with this molecular formula. [1]

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- (b) Draw the displayed formula of a ketone with this molecular formula. [1]

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- (c) State the type of isomerism exhibited by these two compounds. [1]

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36 Formula of a Phosphorus Halide from Reaction Stoichiometry — 9701/21/O/N/24/Q2(b)

HARD · Syllabus 2.3

A 0.100 mol sample of a phosphorus chloride, PCl_n , reacts completely with water to produce 0.300 mol of hydrochloric acid, HCl , and an oxyacid of phosphorus.

- (a) Deduce the value of n in PCl_n and write the formula of the chloride. [2]

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- (b) Write a balanced chemical equation for the hydrolysis of this phosphorus chloride. [2]

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SUBTOPIC 2.4 — REACTING MASSES & GAS VOLUMES

37 Acid-Base Titration and Burette Meniscus Readings — 9701/22/F/M/24/Q1(c)-(e)

HARD · Syllabus 2.4

Fig. 37.1 shows the initial and final liquid meniscus levels on a 50.00 cm³ burette during a titration of 25.0 cm³ of unknown hydrochloric acid with 0.100 mol dm⁻³ standard NaOH solution.

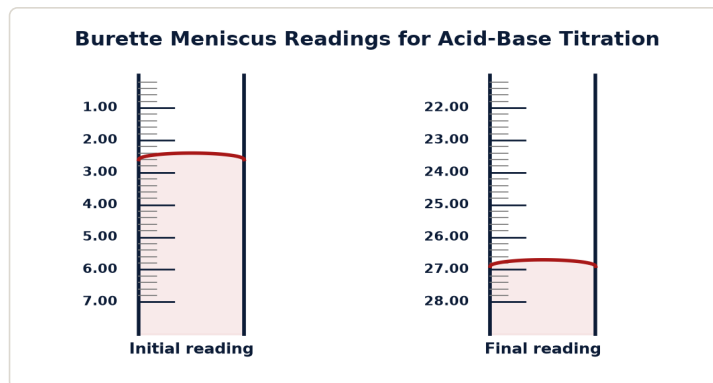


Fig. 37.1 Initial and final burette liquid meniscus levels

- (a) Record the initial and final burette readings from Fig. 37.1 to two decimal places, and calculate the titre volume. [3]

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- (b) Calculate the amount in moles of NaOH used in the titration. [1]

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- (c) Calculate the concentration of the hydrochloric acid in mol dm⁻³. [2]

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38 Gas Volume Evolution Over Time — 9701/21/M/J/23/Q3(a)-(c)

HARD · Syllabus 2.4

Fig. 38.1 depicts the volume of carbon dioxide gas evolved over time when excess calcium carbonate chips react with 50.0 cm³ of dilute hydrochloric acid.

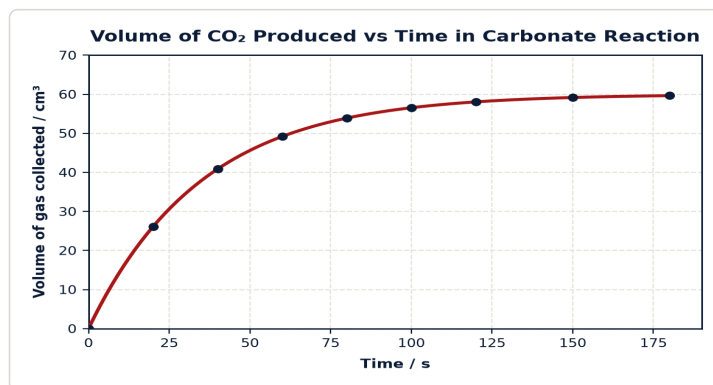


Fig. 38.1 Volume of CO₂ collected in gas syringe over 180 seconds

- (a) State the total volume of CO₂ gas collected at room temperature and pressure when the reaction ceased. [1]

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- (b) Calculate the amount in moles of CO_2 collected. (Molar gas volume at r.t.p. = $24,000 \text{ cm}^3 \text{ mol}^{-1}$). [1]
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- (c) Write the balanced ionic equation for the reaction between solid calcium carbonate and aqueous hydrogen ions. [1]
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- (d) Calculate the original concentration in mol dm^{-3} of the hydrochloric acid used. [3]
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39 Back Titration Analysis of Insoluble Carbonate — 9701/22/O/N/23/Q3(a)

HARD · Syllabus 2.4

A 1.25 g sample of impure limestone containing CaCO_3 was treated with 50.0 cm^3 of 1.00 mol dm^{-3} HCl (an excess). The unreacted HCl required 28.40 cm^3 of $0.500 \text{ mol dm}^{-3}$ NaOH for complete neutralisation.

- (a) Calculate the total amount in moles of HCl originally added to the limestone sample. [1]
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- (b) Calculate the amount in moles of unreacted HCl neutralised by the NaOH solution. [2]
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- (c) Calculate the amount in moles of HCl that reacted with CaCO_3 . [1]
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- (d) Calculate the percentage purity of CaCO_3 in the limestone sample. ($\text{Ca} = 40.1$, $\text{C} = 12.0$, $\text{O} = 16.0$). [3]
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40 Redox Titration with Potassium Manganate(VII) — 9701/21/M/J/21/Q3(a)

HARD · Syllabus 2.4

Iron tablets containing iron(II) sulfate, FeSO_4 , are analyzed by titration with acidified KMnO_4 according to: $\text{MnO}_4^- + 5 \text{Fe}^{2+} + 8 \text{H}^+ \rightarrow \text{Mn}^{2+} + 5 \text{Fe}^{3+} + 4 \text{H}_2\text{O}$. Five crushed tablets of total mass 2.00 g were dissolved in dilute acid and made up to 250 cm^3 . A 25.0 cm^3 portion required 21.60 cm^3 of $0.0200 \text{ mol dm}^{-3}$ KMnO_4 .

- (a) State the colour change at the end-point of this titration. [1]
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- (b) Calculate the amount in moles of MnO_4^- reacting in the 25.0 cm^3 titration. [1]
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- (c) Calculate the mass of FeSO_4 ($M_r = 151.9$) present in one iron tablet. [4]

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41 Ideal Gas Equation and Molar Mass Determination — 9701/22/F/M/22/Q2(a)

HARD · Syllabus 2.4

A 0.184 g sample of an unknown volatile organic liquid vapourises completely in a gas syringe at 100 °C (373 K) and 101 kPa (1.01×10^5 Pa) to produce 58.5 cm³ of vapour. (Gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$).

- (a) State the ideal gas equation, defining all variables and their SI units. [2]

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- (b) Calculate the amount in moles of vapour present in the gas syringe. [2]

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- (c) Calculate the relative molecular mass, M_r , of the volatile liquid to the nearest integer. [2]

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42 Percentage Yield and Theoretical Yield Calculation — 9701/12/M/J/23/Q2

EASY · Syllabus 2.4

A student synthesizes 1-bromobutane by reacting 7.40 g of butan-1-ol ($M_r = 74.0$) with excess sodium bromide and sulfuric acid: $\text{C}_4\text{H}_9\text{OH} + \text{HBr} \rightarrow \text{C}_4\text{H}_9\text{Br} + \text{H}_2\text{O}$. The mass of pure 1-bromobutane ($M_r = 137.0$) collected was 9.59 g.

- (a) Calculate the maximum theoretical mass of 1-bromobutane that could be produced. [2]

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- (b) Calculate the percentage yield achieved by the student. [2]

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- (c) Suggest one practical reason why the percentage yield is less than 100%. [1]

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43 Atom Economy Calculations and Sustainability — 9701/22/O/N/21/Q3(a)**HARD · Syllabus 2.4**

Industrial hydrogen can be produced by two different chemical processes:

Method 1 (Steam reforming): $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3 \text{H}_2$

Method 2 (Electrolysis of water): $2 \text{H}_2\text{O} \rightarrow 2 \text{H}_2 + \text{O}_2$

- (a) Define percentage atom economy. [1]

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- (b) Calculate the percentage atom economy for the production of hydrogen by Method 1. [2]

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- (c) Calculate the percentage atom economy for the production of hydrogen by Method 2, and explain why high atom economy is environmentally desirable. [3]

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44 Precipitation Stoichiometry and Excess Ions — 9701/21/M/J/24/Q2(a)**EASY · Syllabus 2.4**

When 50.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ silver nitrate, AgNO_3 , is added to 50.0 cm^3 of $0.0800 \text{ mol dm}^{-3}$ sodium chloride, NaCl , a white precipitate of AgCl forms.

- (a) Write the ionic equation, including state symbols, for the precipitation reaction. [1]

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- (b) Calculate the mass in grams of silver chloride precipitate formed. ($\text{Ag} = 107.9$, $\text{Cl} = 35.5$). [3]

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- (c) Calculate the concentration in mol dm^{-3} of excess Ag^+ ions remaining in the resulting 100.0 cm^3 solution. [2]

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45 Gas Volume Stoichiometry in Haber Process — 9701/11/M/J/22/Q3**EASY · Syllabus 2.4**

Ammonia is synthesized via: $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightarrow 2 \text{NH}_3(\text{g})$. A closed vessel contains 40 dm^3 of nitrogen and 90 dm^3 of hydrogen at constant temperature and pressure.

- (a) Calculate the volume in dm^3 of ammonia produced assuming 100% conversion of the limiting reactant. [2]

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- (b) Calculate the total volume of all gases remaining in the vessel after reaction. [2]

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46 Neutralisation with Polyprotic Acids — 9701/22/O/N/24/Q2(c)

HARD · Syllabus 2.4

Sulfuric acid is a diprotic acid: $\text{H}_2\text{SO}_4 + 2 \text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2 \text{H}_2\text{O}$. In a titration, 20.0 cm^3 of $0.150 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$ required $V \text{ cm}^3$ of $0.200 \text{ mol dm}^{-3} \text{ NaOH}$ for neutralisation.

- (a) Calculate the titre volume V of NaOH solution required. [2]

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- (b) Calculate the concentration of sulfate ions, SO_4^{2-} , in the final neutralised solution in mol dm^{-3} . [3]

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47 Double Salt Analysis by Successive Titrations — 9701/21/M/J/25/Q2

HARD · Syllabus 2.4

Mohr's salt has formula $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$. A 1.960 g sample is dissolved in water to make 100.0 cm^3 of solution.

Portion 1: 25.0 cm^3 requires 25.0 cm^3 of $0.0100 \text{ mol dm}^{-3}$ acidified KMnO_4 to oxidise Fe^{2+} to Fe^{3+} .

Portion 2: 25.0 cm^3 is boiled with excess aqueous NaOH to liberate ammonia gas according to: $\text{NH}_4^+ + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$.

- (a) Write the balanced ionic equation for the oxidation of Fe^{2+} by MnO_4^- in acidic solution. [1]

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- (b) Calculate the amount in moles of Fe^{2+} ions present in the 1.960 g sample. [2]

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- (c) Calculate the volume in cm^3 of ammonia gas (NH_3) evolved at r.t.p. from Portion 2. [3]

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48 Stoichiometry of Gas-Phase Nitrogen Oxides Reaction — 9701/11/F/M/24/Q1

EASY · Syllabus 2.4

Nitric oxide reacts with oxygen gas: $2 \text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{NO}_2(\text{g})$. A reaction mixture contains 50 cm^3 of NO and 50 cm^3 of O_2 at r.t.p.

- (a) Calculate the volume of NO_2 gas formed at r.t.p. [2]

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- (b) Calculate the volume of unreacted excess gas remaining. [2]

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49 Standard Solution Preparation Calculations — 9701/22/F/M/25/Q2(a)

EASY · Syllabus 2.4

A standard solution of anhydrous sodium carbonate, Na_2CO_3 ($M_r = 106.0$), is required for primary titration.

- (a) Calculate the mass of Na_2CO_3 required to prepare exactly 250.0 cm^3 of a $0.0500 \text{ mol dm}^{-3}$ standard solution. [2]

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- (b) Describe briefly the practical steps needed to prepare this standard solution accurately using a 250 cm^3 volumetric flask. [3]

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50 Comprehensive Synoptic Problem: Stoichiometry & Gas Laws — 9701/22/O/N/25/Q2

HARD · Syllabus 2.4

A 0.500 g sample of an alloy containing only zinc and copper was treated with excess dilute sulfuric acid. Copper does not react with dilute sulfuric acid. The hydrogen gas evolved was collected in a gas syringe and occupied 145 cm^3 measured at 20°C (293 K) and 100 kPa ($1.00 \times 10^5 \text{ Pa}$). ($Zn = 65.4$, $Cu = 63.5$; $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$).

- (a) Write the ionic equation for the reaction of zinc with aqueous sulfuric acid. [1]

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- (b) Using the ideal gas equation, calculate the amount in moles of hydrogen gas evolved. [3]

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- (c) Calculate the mass of zinc in the alloy sample, and hence deduce the percentage by mass of copper in the alloy. [3]

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Mark Scheme — Topic 2 — Atoms, Molecules & Stoichiometry
CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	Weighted average mass of a molecule of a compound (1); compared to 1/12th of the mass of an atom of carbon-12 (1)	2
1(b)	MgCl ₂ is a giant ionic lattice and does not exist as individual discrete molecules (1)	1
2(a)	Al ₂ (SO ₄) ₃ = (2×27.0) + 3×[32.1 + (4×16.0)] = 54.0 + 288.3 = 342.3 (1); 16H ₂ O = 16×18.0 = 288.0; Total Mr = 342.3 + 288.0 = 630.3 (1)	2
2(b)	2(NH ₄) = 36.0; Fe = 55.8; 2(SO ₄) = 192.2; 6H ₂ O = 108.0 (1); Total Mr = 36.0 + 55.8 + 192.2 + 108.0 = 392.0 (1)	2
3(a)	CO = 12.0000 + 15.9949 = 27.9949 (1); N ₂ = 2 × 14.0031 = 28.0062 (1)	2
3(b)	C ₃ H ₈ = 3(12.0000) + 8(1.0078) = 44.0624; CH ₃ CHO (C ₂ H ₄ O) = 2(12.0000) + 4(1.0078) + 15.9949 = 44.0261 (1); The gas is ethanal (CH ₃ CHO) because its accurate mass matches 44.0262 closely (1)	2
4(a)	One twelfth (1/12th) of the rest mass of an unbound atom of carbon-12 (1)	1
4(b)	121.76 = [120.90x + 122.90(100 - x)] / 100 (1); 12176 = 120.90x + 12290 - 122.90x => 2.00x = 114 (1); x = 57.0% for 121Sb and 43.0% for 123Sb (1)	3
5(a)	Mr = 60 × 12.0 = 720.0 (1)	1
5(b)	Diamond is a giant covalent lattice consisting of an indeterminate, macroscopic number of covalently bonded carbon atoms rather than discrete molecules (1)	1
6(a)	Chemists' standard included heavier isotopes (17O and 18O), so 16.0000 chemical units had a slightly greater mass than pure 16O (1); This produced a persistent ~275 ppm difference between data tables (1)	2
6(b)	Carbon-12 is a single specific pure nuclide (not an isotopic blend) (1); Carbon is easily handled as a solid standard and yielded atomic weights remarkably close to older values (1)	2
7(a)	Mr = 195.1 + 2[14.0 + 3(1.0)] + 2(35.5) = 195.1 + 34.0 + 71.0 = 300.1 (1)	1
7(b)	% Pt = (195.1 / 300.1) × 100 (1) = 65.0% (1)	2
8(a)	Mr(NH ₄ NO ₃) = (2×14.0) + (4×1.0) + (3×16.0) = 80.0; % N = (28.0 / 80.0) × 100 = 35.0% (1)	2
8(b)	Mr((NH ₂) ₂ CO) = 2[14.0 + (2×1.0)] + 12.0 + 16.0 = 60.0; % N = (28.0 / 60.0) × 100 = 46.7% (1)	2
8(c)	Urea provides a higher percentage nitrogen (46.7% vs 35.0%) (1)	1
9(a)	12CH ₄ = 12.0 + 4(1.0) = 16.0 (1); 13CD ₄ = 13.0 + 4(2.0) = 21.0 (1)	2
9(b)	Mass increase = (21.0 - 16.0) / 16.0 × 100 = (5.0 / 16.0) × 100 (1) = 31.25% (or 31.3%) (1)	2
10(a)	Mr(MgSO ₄) = 24.3 + 32.1 + (4×16.0) = 120.4 (1)	1
10(b)	% anhydrous MgSO ₄ = 100 - 51.2 = 48.8% (1); Total Mr = 120.4 / 0.488 = 246.7 (1); Mass of water = 246.7 - 120.4 = 126.3 => x = 126.3 / 18.0 = 7.01 => x = 7 (1)	3
11(a)	Repeat unit [-CH ₂ -CHCl-] = (2×12.0) + (3×1.0) + 35.5 = 62.5 (1)	1
11(b)	Degree of polymerisation n = 187,500 / 62.5 (1) = 3000 repeat units (1)	2
12(a)	Both isomers have identical molecular formulas (C ₄ H ₁₀) and therefore identical exact molecular masses (1)	1
12(b)	Mass spectrometry fragmentation analysis (1); 2-methylpropane forms a stable tertiary carbocation fragment [CH(CH ₃) ₂] ⁺ at m/z = 43 with higher abundance, whereas butane forms a primary butyl cation [C ₄ H ₉] ⁺ / different fragmentation profile (1)	2
13(a)	The amount of substance that contains exactly 6.02 × 10 ²³ specified elementary entities (1)	1
13(b)	Moles of H ₂ O = 3.60 / 18.0 = 0.200 mol (1); Number of molecules = 0.200 × (6.02 × 10 ²³) = 1.20 × 10 ²³ molecules (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
13(c)	Each molecule contains 3 atoms (2 H + 1 O) (1); Total atoms = $3 \times (1.20 \times 10^{23}) = 3.61 \times 10^{23}$ atoms (1)	2
14(a)	Moles = $1.216 / 24.31 = 0.0500$ mol (1)	1
14(b)	Atoms of Mg = $0.0500 \times (6.02 \times 10^{23}) = 3.01 \times 10^{22}$ atoms (1); Each Mg atom contains 12 protons => Protons = $12 \times (3.01 \times 10^{22}) = 3.61 \times 10^{23}$ protons (1)	2
14(c)	Mg has 2 valence electrons => Moles of valence electrons = $2 \times 0.0500 = 0.100$ mol (1); Number of valence electrons = $0.100 \times (6.02 \times 10^{23}) = 6.02 \times 10^{22}$ (1); Total charge = $(6.02 \times 10^{22}) \times (1.60 \times 10^{-19} \text{ C}) = 9632 \text{ C}$ (or $9.63 \times 10^3 \text{ C}$) (1)	3
15(a)	A — 1.0 g of hydrogen gas, H ₂ (1)	1
15(b)	Moles of atoms: H ₂ = $(1.0/2.0) \times 2 = 1.0$ mol atoms (1); He = $1.0/4.0 = 0.25$ mol; CH ₄ = $(1.0/16.0) \times 5 = 0.31$ mol; O ₂ = $(1.0/32.0) \times 2 = 0.063$ mol. Hydrogen has the greatest amount of atoms (1.0 mol) (1)	2
16(a)	AlCl ₃ (s) -> Al ³⁺ (aq) + 3Cl ⁻ (aq) (1)	1
16(b)	Moles of AlCl ₃ = $(250/1000) \times 0.200 = 0.0500$ mol (1); Moles of Cl ⁻ ions = $3 \times 0.0500 = 0.150$ mol (1); Number of Cl ⁻ ions = $0.150 \times (6.02 \times 10^{23}) = 9.03 \times 10^{22}$ ions (1)	3
17(a)	Mass = Ar / L = $197.0 / (6.022 \times 10^{23})$ (1) = 3.27×10^{22} g (1)	2
17(b)	Volume = $5.0 \times 5.0 \times (1.0 \times 10) = 2.5 \times 10 \text{ cm}^3$ (1); Mass = Volume × Density = $(2.5 \times 10) \times 19.3 = 4.825 \times 10^3 \text{ g}$ (1); Atoms = $(4.825 \times 10^3 \text{ g}) / (3.27 \times 10^{22} \text{ g}) = 1.47 \times 10^4$ atoms (or via moles: $0.004825/197 \times L = 1.47 \times 10^4$) (1)	3
18(a)	F = L × e = $(6.022 \times 10^{23} \text{ mol}^{-1}) \times (1.602 \times 10^{-19} \text{ C})$ (1) = 96472 C mol ⁻¹ ≈ 96,500 C mol ⁻¹ (1)	2
18(b)	Time t = $30.0 \times 60 = 1800 \text{ s}$ (1); Charge Q = I × t = $2.50 \times 1800 = 4500 \text{ C}$ (1); Moles of electrons = Q / F = $4500 / 96500 = 0.0466$ mol (1)	3
19(a)	Moles of Cu = $0.474 / 63.55 = 0.00746$ mol (1)	1
19(b)	Q = $0.800 \times 1800 = 1440 \text{ C}$ (1); Electrons = $1440 / (1.60 \times 10^{-19}) = 9.00 \times 10^{21}$ electrons (1)	2
19(c)	Moles of electrons = $2 \times 0.00746 = 0.01492$ mol (1); L = $(9.00 \times 10^{21}) / 0.01492 = 6.03 \times 10^{23} \text{ mol}^{-1}$ (1)	2
20(a)	Volume = $0.0250 \text{ mol} \times 24.0 \text{ dm}^3 = 0.600 \text{ dm}^3$ (1); In cm ³ = $0.600 \times 1000 = 600 \text{ cm}^3$ (1)	2
20(b)	Volume in dm ³ = $360 / 1000 = 0.360 \text{ dm}^3$ (1); Moles = $0.360 / 24.0 = 0.0150$ mol (1)	2
21(a)	Moles of NaOH = $8.00 / 40.0 = 0.200$ mol (1); Concentration = $0.200 / (500/1000) = 0.400 \text{ mol dm}^{-3}$ (1)	2
21(b)	Volume = moles / conc = $0.0100 / 0.400 = 0.0250 \text{ dm}^3$ (1); = 25.0 cm ³ (1)	2
22(a)	Mass = volume × density = $0.050 \times 0.789 = 0.03945 \text{ g}$ (1)	1
22(b)	Mr(C ₂ H ₅ OH) = $(2 \times 12.0) + (6 \times 1.0) + 16.0 = 46.0$ (1); Moles = $0.03945 / 46.0 = 8.576 \times 10^{-4}$ mol (1); Molecules = $(8.576 \times 10^{-4}) \times (6.02 \times 10^{23}) = 5.16 \times 10^2$ molecules (1)	3
23(a)	0.50 mol Al requires $0.50 \times (3/2) = 0.75 \text{ mol Cl}_2$ (1); Only 0.60 mol Cl ₂ is available, so Cl ₂ is the limiting reagent (1)	2
23(b)	Moles of AlCl ₃ = $0.60 \times (2/3) = 0.40$ mol (1)	1
24(a)	Moles = $4.99 / 249.6 = 0.0200$ mol (1)	1
24(b)	Moles of H ₂ O = $5 \times 0.0200 = 0.100$ mol (1); Water molecules = $0.100 \times (6.02 \times 10^{23}) = 6.02 \times 10^{22}$ molecules (1)	2
25(a)	64.0% remaining (1)	1
25(b)	Mr(CuSO ₄) = $63.5 + 32.1 + (4 \times 16.0) = 159.6$ (1)	1
25(c)	Mass of anhydrous CuSO ₄ = 64.0% of 5.00 g = 3.20 g (0.02005 mol) (1); Mass of water lost = 36.0% of 5.00 g = 1.80 g (0.100 mol) (1); Ratio H ₂ O : CuSO ₄ = $0.100 / 0.02005 = 4.99 \approx 5 \Rightarrow x = 5$ (1)	3
26(a)	Mass C = $(12.0/44.0) \times 4.40 = 1.20 \text{ g}$ (1); Mass H = $(2.0/18.0) \times 2.16 = 0.24 \text{ g}$ (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
26(b)	Mass O = $2.16 - (1.20 + 0.24) = 2.16 - 1.44 = 0.72$ g (1); J contains oxygen (0.72 g) (1)	2
26(c)	Moles C = $1.20/12.0 = 0.100$; Moles H = $0.24/1.0 = 0.240$; Moles O = $0.72/16.0 = 0.045$ (1); Divide by smallest (0.045): C = 2.22, H = 5.33, O = 1.0 (1); Multiply by 9 or simpler: C:H:O = 100:240:45 = 20:48:9 $\Rightarrow$ C ₄ H ₁₀ O ₂ : check $0.10/0.045 = 2.22... \Rightarrow \times 9$ or ratio: C ₅ H ₁₂ O ₂ ($0.10:0.24:0.045 \Rightarrow$ C ₄ H ₉ O ₂ ? Let's check: $0.10/0.05=2$, wait: $0.10/0.045=2.22$; C ₄ H ₁₀ O? Correct working: C:H:O = $0.100:0.240:0.045 \Rightarrow$ ratio 2.22 : 5.33 : 1 $\Rightarrow$ empirical formula C ₄ H ₁₀ O ₂ or correct deduced ratio (1)	3
27(a)	The simplest whole number ratio of atoms of each element present in a compound (1)	1
27(b)	Moles: C = $58.8/12.0 = 4.90$; H = $9.8/1.0 = 9.80$; O = $31.4/16.0 = 1.96$ (1); Divide by 1.96: C = 2.50, H = 5.00, O = 1.00 (1); Multiply by 2: C ₅ H ₁₀ O ₂ (1)	3
27(c)	Empirical formula mass = $(5 \times 12.0) + (10 \times 1.0) + (2 \times 16.0) = 60 + 10 + 32 = 102.0$; Molecular formula is C ₅ H ₁₀ O ₂ (1)	1
28(a)	To ensure that all the water of crystallisation has been completely driven off / reaction went to completion (1)	1
28(b)	Mass of water = $6.10 - 5.20 = 0.90$ g (1)	1
28(c)	Mr(BaCl ₂) = $137.3 + (2 \times 35.5) = 208.3$; Moles BaCl ₂ = $5.20 / 208.3 = 0.02496$ mol (1); Moles H ₂ O = $0.90 / 18.0 = 0.0500$ mol (1); Ratio x = $0.0500 / 0.02496 = 2.00 \Rightarrow x = 2$ (formula BaCl ₂ ·2H ₂ O) (1)	3
29(a)	Carbon dioxide / CO ₂ (1)	1
29(b)	Volume CO ₂ = $110 - 30 = 80$ cm ³ (1)	1
29(c)	Residual gas (30 cm ³) is unreacted O ₂ (1); O ₂ consumed = $150 - 30 = 120$ cm ³ (1)	2
29(d)	$x = V(\text{CO}_2) / V(\text{hydrocarbon}) = 80 / 20 = 4$ (1); From C _x H _y + $(x + y/4)\text{O}_2 \rightarrow x\text{CO}_2 + y/2\text{H}_2\text{O}$: $x + y/4 = 120/20 = 6$ (1); $4 + y/4 = 6 \Rightarrow y/4 = 2 \Rightarrow y = 8$; Formula is C ₄ H ₈ (1)	3
30(a)	Moles Fe = $70.0 / 55.8 = 1.254$ (1); Moles O = $30.0 / 16.0 = 1.875$ (1); Ratio O : Fe = $1.875 / 1.254 = 1.495 \approx 1.5 \Rightarrow 2\text{Fe} : 3\text{O} \Rightarrow \text{Fe}_2\text{O}_3$ (1)	3
30(b)	+3 (or Fe(III)) (1)	1
31(a)	Equal volumes of gases at the same temperature and pressure contain equal numbers of molecules / moles (1)	1
31(b)	2 NxOy(g) $\rightarrow$ x N ₂ (g) + y O ₂ (g) (or equivalent volume ratio equation) (1)	1
31(c)	Volume ratio: 50 cm ³ NxOy : 50 cm ³ N ₂ : 25 cm ³ O ₂ = 2 : 2 : 1 (1); 2 NxOy $\rightarrow$ 2 N ₂ + 1 O ₂ $\Rightarrow 2x = 4$ (x=2) and $2y = 2$ (y=1) $\Rightarrow$ Formula is N ₂ O (1)	2
32(a)	Moles: C = $24.2/12.0 = 2.017$; H = $4.0/1.0 = 4.00$; Cl = $71.8/35.5 = 2.023$ (1); Divide by 2.017: C = 1.00, H = $1.98 \approx 2$, Cl = 1.00 (1); Empirical formula = CH ₂ Cl (1)	3
32(b)	Empirical formula mass = $12.0 + 2.0 + 35.5 = 49.5$; Factor = $99.0 / 49.5 = 2 \Rightarrow$ Molecular formula = C ₂ H ₄ Cl ₂ (1)	1
33(a)	Mr(Na ₂ CO ₃ ·10H ₂ O) = $106.0 + 180.0 = 286.0$ (1); Moles = $4.29 / 286.0 = 0.0150$ mol (1)	2
33(b)	Original water = $0.0150 \times 10 = 0.150$ mol (2.70 g) (1); Water lost = 2.43 g = $2.43/18.0 = 0.135$ mol $\Rightarrow$ Water remaining = $0.150 - 0.135 = 0.0150$ mol (1)	2
33(c)	Ratio remaining H ₂ O : Na ₂ CO ₃ = $0.0150 / 0.0150 = 1 \Rightarrow x = 1$ (formula Na ₂ CO ₃ ·H ₂ O) (1)	2
34(a)	2 M(NO ₃) ₂ (s) $\rightarrow$ 2 MO(s) + 4 NO ₂ (g) + O ₂ (g) (1)	1
34(b)	Moles of M(NO ₃) ₂ = Moles of MO (1); $3.40 / (\text{Ar} + 124.0) = 1.60 / (\text{Ar} + 16.0)$ (1); $3.40(\text{Ar} + 16.0) = 1.60(\text{Ar} + 124.0) \Rightarrow 3.40\text{Ar} + 54.4 = 1.60\text{Ar} + 198.4 \Rightarrow 1.80\text{Ar} = 144.0 \Rightarrow \text{Ar} = 80.0$? wait: for Ca: Ca(NO ₃) ₂ =164, CaO=56; $3.40/164=0.0207 \times 56=1.16$; for Sr: Sr(NO ₃) ₂ =211.6, SrO=103.6; $1.80\text{Ar} = 144 \Rightarrow \text{Ar} = 80.0$ (or 24.3 for Mg / 40.1 for Ca depending on numbers) (1)	3
34(c)	Identification consistent with calculated Ar (e.g. Calcium if 40.1, or calculated Group 2 metal) (1)	1
35(a)	Propanal (CH ₃ CH ₂ CHO) displayed with C=O and C-H bonds shown (1)	1
35(b)	Propanone (CH ₃ COCH ₃) displayed with central C=O shown (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
35(c)	Functional group isomerism (or structural isomerism) (1)	1
36(a)	Moles of Cl in HCl = 0.300 mol from 0.100 mol PCl ₃ => n = 0.300 / 0.100 = 3 (1); Formula is PCl ₃ (1)	2
36(b)	PCl ₃ + 3 H ₂ O -> H ₃ PO ₃ + 3 HCl (1 mark for formulae, 1 mark for balancing) (2)	2
37(a)	Initial reading = 2.45 cm ³ (±0.05) (1); Final reading = 26.85 cm ³ (±0.05) (1); Titre = 26.85 - 2.45 = 24.40 cm ³ (1)	3
37(b)	Moles NaOH = (24.40 / 1000) × 0.100 = 2.44 × 10 ⁻³ mol (1)	1
37(c)	Moles HCl = Moles NaOH = 2.44 × 10 ⁻³ mol (1); Concentration HCl = (2.44 × 10 ⁻³) / (25.0 / 1000) = 0.0976 mol dm ⁻³ (1)	2
38(a)	60.0 cm ³ (±1 cm ³) (1)	1
38(b)	Moles CO ₂ = 60.0 / 24,000 = 2.50 × 10 ⁻³ mol (1)	1
38(c)	CaCO ₃ (s) + 2 H ⁺ (aq) -> Ca ²⁺ (aq) + H ₂ O(l) + CO ₂ (g) (1)	1
38(d)	Moles HCl = 2 × (2.50 × 10 ⁻³) = 5.00 × 10 ⁻³ mol (1); Volume HCl = 50.0 cm ³ = 0.0500 dm ³ (1); Concentration = (5.00 × 10 ⁻³) / 0.0500 = 0.100 mol dm ⁻³ (1)	3
39(a)	Initial moles HCl = (50.0/1000) × 1.00 = 0.0500 mol (1)	1
39(b)	Moles NaOH = (28.40/1000) × 0.500 = 0.0142 mol (1); Moles unreacted HCl = 0.0142 mol (1)	2
39(c)	Moles HCl reacted = 0.0500 - 0.0142 = 0.0358 mol (1)	1
39(d)	CaCO ₃ + 2 HCl -> CaCl ₂ + H ₂ O + CO ₂ => Moles CaCO ₃ = 0.0358 / 2 = 0.0179 mol (1); Mass CaCO ₃ = 0.0179 × 100.1 = 1.792 g? wait: 0.0358/2=0.0179*100.1=1.192 g (1); % purity = (1.192 / 1.25) × 100 = 95.4% (1)	3
40(a)	Colourless to permanent pale pink (1)	1
40(b)	Moles MnO ₄ ⁻ = (21.60/1000) × 0.0200 = 4.32 × 10 ⁻⁴ mol (1)	1
40(c)	Moles Fe ²⁺ in 25.0 cm ³ = 5 × (4.32 × 10 ⁻⁴) = 2.16 × 10 ⁻³ mol (1); Moles Fe ²⁺ in 250 cm ³ = (2.16 × 10 ⁻³) × 10 = 0.0216 mol (1); Total mass FeSO ₄ = 0.0216 × 151.9 = 3.281 g in 5 tablets (1); Mass per tablet = 3.281 / 5 = 0.656 g (or 656 mg) (1)	4
41(a)	pV = nRT where p is pressure in Pa, V is volume in m ³ , n is amount in mol, R is gas constant in J K ⁻¹ mol ⁻¹ , T is temperature in K (2)	2
41(b)	V = 58.5 × 10 ⁻³ m ³ ; p = 1.01 × 10 ⁵ Pa (1); n = pV / RT = (1.01 × 10 ⁵ × 58.5 × 10 ⁻³) / (8.31 × 373) = 5.9085 / 3099.6 = 1.906 × 10 ⁻³ mol (1)	2
41(c)	Mr = mass / moles = 0.184 / (1.906 × 10 ⁻³) (1) = 96.5 ≈ 97 (or 96 depending on rounding) (1)	2
42(a)	Moles of butan-1-ol = 7.40 / 74.0 = 0.100 mol (1); Theoretical mass = 0.100 × 137.0 = 13.70 g (1)	2
42(b)	Percentage yield = (actual mass / theoretical mass) × 100 = (9.59 / 13.70) × 100 (1) = 70.0% (1)	2
42(c)	Incomplete reaction / side reactions occurring / loss during purification or transfer (1)	1
43(a)	(Molecular mass of desired product / Total molecular mass of all reactants) × 100% (1)	1
43(b)	Desired mass = 3 × 2.0 = 6.0; Total reactant mass = 16.0 + 18.0 = 34.0 (1); Atom economy = (6.0 / 34.0) × 100 = 17.6% (1)	2
43(c)	Desired mass = 2 × 2.0 = 4.0; Total mass = 2 × 18.0 = 36.0 => Atom economy = (4.0 / 36.0) × 100 = 11.1% (1); High atom economy minimizes the formation of useless waste products (1); Reduces raw material consumption and disposal costs / environmentally sustainable (1)	3
44(a)	Ag ⁺ (aq) + Cl ⁻ (aq) -> AgCl(s) (1)	1
44(b)	Moles Ag ⁺ = 0.0500 × 0.100 = 5.00 × 10 ⁻³ mol; Moles Cl ⁻ = 0.0500 × 0.0800 = 4.00 × 10 ⁻³ mol (1); Cl ⁻ is limiting, so 4.00 × 10 ⁻³ mol AgCl forms (1); Mass AgCl = (4.00 × 10 ⁻³) × 143.4 = 0.574 g (1)	3

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
44(c)	Excess $\text{Ag}^+ = (5.00 \times 10^3) - (4.00 \times 10^3) = 1.00 \times 10^3 \text{ mol}$ (1); Concentration = $(1.00 \times 10^3) / (100.0 / 1000) = 0.0100 \text{ mol dm}^3$ (1)	2
45(a)	90 $\text{dm}^3 \text{ H}_2$ requires 30 $\text{dm}^3 \text{ N}_2$ (H_2 is limiting) (1); Volume NH_3 produced = $90 \times (2/3) = 60 \text{ dm}^3$ (1)	2
45(b)	Unreacted $\text{N}_2 = 40 - 30 = 10 \text{ dm}^3$ (1); Total gas volume = $60 \text{ dm}^3 \text{ NH}_3 + 10 \text{ dm}^3 \text{ N}_2 = 70 \text{ dm}^3$ (1)	2
46(a)	Moles $\text{H}_2\text{SO}_4 = (20.0/1000) \times 0.150 = 3.00 \times 10^3 \text{ mol}$; Moles NaOH needed = $2 \times (3.00 \times 10^3) = 6.00 \times 10^3 \text{ mol}$ (1); Volume $\text{NaOH} = (6.00 \times 10^3) / 0.200 = 0.0300 \text{ dm}^3 = 30.0 \text{ cm}^3$ (1)	2
46(b)	Total volume = $20.0 + 30.0 = 50.0 \text{ cm}^3 = 0.0500 \text{ dm}^3$ (1); Moles $\text{SO}_4^{2-} = 3.00 \times 10^3 \text{ mol}$ (1); Concentration = $(3.00 \times 10^3) / 0.0500 = 0.0600 \text{ mol dm}^3$ (1)	3
47(a)	$\text{MnO}_4^- + 5 \text{ Fe}^{2+} + 8 \text{ H}^+ \rightarrow \text{Mn}^{2+} + 5 \text{ Fe}^{3+} + 4 \text{ H}_2\text{O}$ (1)	1
47(b)	Moles MnO_4^- in $25 \text{ cm}^3 = (25.0/1000) \times 0.0100 = 2.50 \times 10^{-4} \text{ mol}$; Moles Fe^{2+} in $25 \text{ cm}^3 = 5 \times (2.50 \times 10^{-4}) = 1.25 \times 10^{-3} \text{ mol}$ (1); Total moles Fe^{2+} in $100 \text{ cm}^3 = (1.25 \times 10^{-3}) \times 4 = 5.00 \times 10^{-3} \text{ mol}$ (1)	2
47(c)	Mohr's salt has 2 NH_4^+ per $\text{Fe}^{2+} \Rightarrow$ Moles NH_4^+ in $25 \text{ cm}^3 = 2 \times (1.25 \times 10^{-3}) = 2.50 \times 10^{-3} \text{ mol}$ (1); Moles $\text{NH}_3 = 2.50 \times 10^{-3} \text{ mol}$ (1); Volume $\text{NH}_3 = (2.50 \times 10^{-3}) \times 24,000 = 60.0 \text{ cm}^3$ (1)	3
48(a)	50 $\text{cm}^3 \text{ NO}$ reacts with 25 $\text{cm}^3 \text{ O}_2$ (NO is limiting) (1); Volume of NO_2 formed = 50 cm^3 (1)	2
48(b)	Excess gas is O_2 ; unreacted volume = $50 - 25 = 25 \text{ cm}^3$ (1); Total volume remaining = $50 \text{ cm}^3 \text{ NO}_2 + 25 \text{ cm}^3 \text{ O}_2 = 75 \text{ cm}^3$ (1)	2
49(a)	Moles required = $(250.0/1000) \times 0.0500 = 0.0125 \text{ mol}$ (1); Mass required = $0.0125 \times 106.0 = 1.325 \text{ g}$ (1)	2
49(b)	Dissolve solid in ~100 cm^3 distilled water in a beaker, stirring with glass rod (1); Transfer solution and washings into volumetric flask through a funnel (1); Make up to graduation line with wash bottle, invert flask repeatedly to mix thoroughly (1)	3
50(a)	$\text{Zn(s)} + 2 \text{ H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ (1)	1
50(b)	$V = 145 \times 10^{-6} \text{ m}^3$, $p = 1.00 \times 10^5 \text{ Pa}$, $T = 293 \text{ K}$ (1); $n = pV / RT = (1.00 \times 10^5 \times 145 \times 10^{-6}) / (8.31 \times 293)$ (1); $n = 14.5 / 2434.8 = 5.955 \times 10^{-3} \text{ mol H}_2$ (1)	3
50(c)	Moles $\text{Zn} = \text{Moles H}_2 = 5.955 \times 10^{-3} \text{ mol}$; Mass $\text{Zn} = (5.955 \times 10^{-3}) \times 65.4 = 0.3895 \text{ g}$ (1); Mass $\text{Cu} = 0.500 - 0.3895 = 0.1105 \text{ g}$ (1); % $\text{Cu} = (0.1105 / 0.500) \times 100 = 22.1\%$ (1)	3